# ETL with Python and SQL

## Data engineering · Level 4 · Lesson 2



# In this lesson you will

- Build extract, transform, load.
- Make loads idempotent: delete-then-insert, or upsert.
- Load incrementally and log every run.

# Key ideas

- A rerun must give the same answer.
- Save the watermark after the load.
- Retry passing faults; set bad data aside.

# Break the pipeline

1. Draw (4 min): ETL for attendance
2. Fail (5 min): stop it halfway
3. Fix (5 min): make it idempotent
4. Log (3 min): what would tell you?

**The lab's numbers (84 of 90, and 152 after a bad retry) make a good worked example.**

# Let’s discuss

1. Why must the watermark be saved after the load?
2. When would you choose upsert over delete-then-insert?
3. Which failures should never be retried?

# Exit ticket

- Write the two SQL statements of an idempotent daily load.